

AARON MULANDI

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SUMMARY

Final-year Electronic and Computer Engineering student with production web development experience and a background in electronics and embedded systems. Designed and built an award-winning offline-first education platform spanning Linux edge servers, delay-tolerant networking, and kernel-level traffic shaping. B.Sc. expected December 2026. Authorized to work in the US without sponsorship.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C, Bash, SQL

Backend & Systems: Node.js, REST APIs, PostgreSQL, Linux (Ubuntu), Docker, Git, networking (TCP/IP, routing, QoS), Linux traffic control (HTB, SFQ)

Frontend & Hardware: React, Next.js, HTML/CSS, GSAP, Three.js; Raspberry Pi, GSM/SMS modules, instrumentation and control systems

Currently learning: PyTorch, distributed training (DDP/FSDP), CUDA/Triton, Kubernetes

EXPERIENCE

Founder & Software Engineer — EduAccess (Independent Project)

2025 – Present

Remote — New Jersey, US / Kenya

- Awarded 2nd runner-up, Engineers Board of Kenya Engineering Recognition & Excellence Awards 2026.
- Built an offline-first educational platform delivering full curriculum content to schools with no reliable internet, architected around a Kolibri LMS edge server on Ubuntu 24.04 over a local network.
- Built a Raspberry Pi Zero 2W delay-tolerant networking node that store-and-forwards content updates between connected and disconnected sites, removing any dependency on permanent connectivity.
- Implemented HTB + SFQ traffic shaping achieving a Jain's Fairness Index above 0.95 and sub-50ms latency under 30-device classroom load, with bandwidth tiers enforced per payment package.
- Integrated M-Pesa STK Push payments with GSM/SMS confirmation, enabling community-managed access for users with no data connection.

Web Developer — Noah's Navy Communications

Mar 2023 – Jul 2024

Nairobi, Kenya

- Developed and maintained production web applications in JavaScript, TypeScript, React, and Next.js, owning features end to end from design handoff through release.
- Collaborated with cross-functional design and product teams to implement interfaces to UI/UX specification across client projects.
- Optimized application performance using GSAP and Three.js for animation and 3D rendering.

Engineering Attachment, Instrumentation & Control — KenGen PLC

Jan 2023 – Apr 2023

Olkaria II Geothermal Power Plant, Naivasha, Kenya

- Specified machine components against electrical requirements and identified procurement inefficiencies in the plant parts pipeline, reducing downtime costs.

Data Annotator, Computer Vision — Sama

Aug 2022 – Oct 2022

Nairobi, Kenya

- Annotated radar and lidar datasets for autonomous vehicle model training, sustaining a 97% accuracy rate against quality benchmarks.

Engineering Attachment, Electrical — KenGen PLC

Oct 2019 – Dec 2019

Gitaru Hydroelectric Power Plant, Embu, Kenya

- Conducted an energy audit that identified reallocation opportunities, contributing to a 10% reduction in plant expenses.

EDUCATION

B.Sc., Electronic and Computer Engineering

2017 – Dec 2026 (expected)

Jomo Kenyatta University of Agriculture and Technology (JKUAT), Kenya

- Final-year project: EduAccess — offline-first educational access system (see above).
- Coursework: computer architecture, embedded systems, DSP, control systems, algorithms, computer networks.

ADDITIONAL EXPERIENCE

Founder & Operator — Munchpad Bakery

Oct 2023 – Present

- Founded and operate a profitable retail bakery; manage sourcing, production, pricing, and marketing.

Server — Applebee's Neighborhood Grill + Bar

Aug 2024 – Sep 2025

- Served guests in a high-volume restaurant, managing multiple tables, orders, and payments under time pressure.